

Mark "Xavier" Bonavita

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<https://xvr6.dev/>

Education

Worcester Polytechnic Institute (WPI) | Worcester, MA

GPA 3.60 / 4.0

- **Masters of Science** in Computer Science

08/2025 - 05/2026

- **Bachelor of Science** in Computer Science

08/2022 - 10/2025

Technical Skills

Languages	C, C++, Go, HTML/CSS, Java, JavaScript/TypeScript, JSON, Python, SQL,
Tools/Frameworks	AWS, Azure, Docker, Flask, Git, REST, React, Unix, Nix, SQLite, PostgreSQL, Vue

Professional Experience

Field-Deployable DNA Classification System (NSF-Sponsored) | Worcester, MA

Mar - Oct 2025

Lead Machine Learning Engineer & UX Consultant

- Led an NSF-funded project building a field-ready DNA species identification pipeline using metric-based analysis and a ViT (Vision Transformer) model on DNA denaturation patterns.
- Achieved 95.35% accuracy by applying Optuna hyperparameter tuning and generating synthetic data to correct dataset imbalance within a PyTorch workflow.
- Built a React Native app enabling real-time species identification through a remote classification server.
- Co-authored a research publication demonstrating applications in biodiversity protection, conservation biology, and anti-wildlife-trafficking efforts.

Projects

Recruit.me, Full-Stack Job Matching & Hiring Platform | Worcester, MA

Aug - Dec 2025

- Built a full-stack recruitment platform using React, React Router, and a RESTful API backed by AWS Lambda and serverless RDS (MySQL), enabling responsive UIs and cost-efficient backend operations.
- Engineered the backend with normalized schemas and scalable query patterns to support jobs, applications, user accounts, and company management workflows.
- Implemented Lambda endpoints in a REST API to support job listings, applicant status changes, company actions, and admin reporting with low-latency communication.
- Developed multi-user workflows (company / applicant / admin), added backend pagination for large datasets, and produced wireframes, UML diagrams, and technical documentation used for team collaboration.

furspace.blog, Humanities Practicum | Worcester, MA

Jul '24 - Mar 2025

- Designed and implemented a nostalgic early-2000s-style storytelling website, taking inspiration from platforms like MySpace to create a distinct digital retro atmosphere.
- Built an interactive reading experience with a continuous scrolling narrative supported by blog-style layout transitions.
- Integrated tooling to automate display of artwork, video, and multimedia content.
- Contributed to the creative direction and visual identity, ensuring cohesive artistic style across the platform.

When2Goat, Meeting and Schedule Coordination Website | Worcester, MA

Mar - May 2024

- Led a four-member team to build a multi-device scheduling application in 5 weeks using Agile/Scrum.
- Built a scalable Flask (Python) backend hosted on Azure with PostgreSQL.
- Improved user experience with a simplified feature-rich interface that remained consistent across device sizes and screen formats.
- Integrated Microsoft Sign-On (MS Entra ID) for secure identity verification and meeting coordination.